BBK3CW-Z results for NH_3 vs experiment

El Mir et al., J. Phys. B 54, 015201 (2020) — E_s = 500 eV, $\theta_s = -6°$, E_e = 74 eV, coplanar asymmetric geometry. Orbitals $3a_1$, 1e, $2a_1$ (Moccia one-centre SCF wave functions).

How to read the figures

- Every model curve is scaled to the maximum of the experimental binary peak. The comparison is therefore of **shapes and binary/recoil ratios**, not of absolute magnitudes.
- The experiment has no points between 160° and 200°, the angular range the detectors cannot reach. The experimental recoil maximum is taken in 200–335°.
- The momentum transfer **K** points at $\theta_K \approx 50°$ (46° for $2a_1$), so $-$**K** is at ≈ 230°. In a first-order picture the binary peak lies near **K** and the recoil peak near $-$**K**.
- Three versions of the model are shown: the original BBK3CW-Z (Fig. 1); with the short-range projectile–core potential, option isr = 1 (Fig. 2); and with the ejected wave orthogonalized to the orbital, option iorth = 1, with and without isr (Fig. 3).

Summary of the numbers

Orbital / version	Binary peak	Recoil peak	Binary / recoil
$3a_1$ experiment	54°	242°	2.1
BBK3CW-Z	60°	flat (max 198°)	14.8
+ short-range (isr = 1)	66°	252°	0.98
orthogonalized	54°	none	30
1e experiment	49°	240°	2.5
BBK3CW-Z	54°	flat (max 192°)	23.4
+ short-range	60°	broad plateau	1.8
orthogonalized	54°	none	37
$2a_1$ experiment	50°	230°	1.1
BBK3CW-Z	48°	246° (weak)	5.3
+ short-range	66° (deformed)	258°	0.7
orthogonalized	42°	none	3.0

“none”: for the orthogonalized runs the largest value between 180° and 340° is at 336°, on the tail of the binary peak; there is no real recoil peak.

1. Original BBK3CW-Z vs experiment

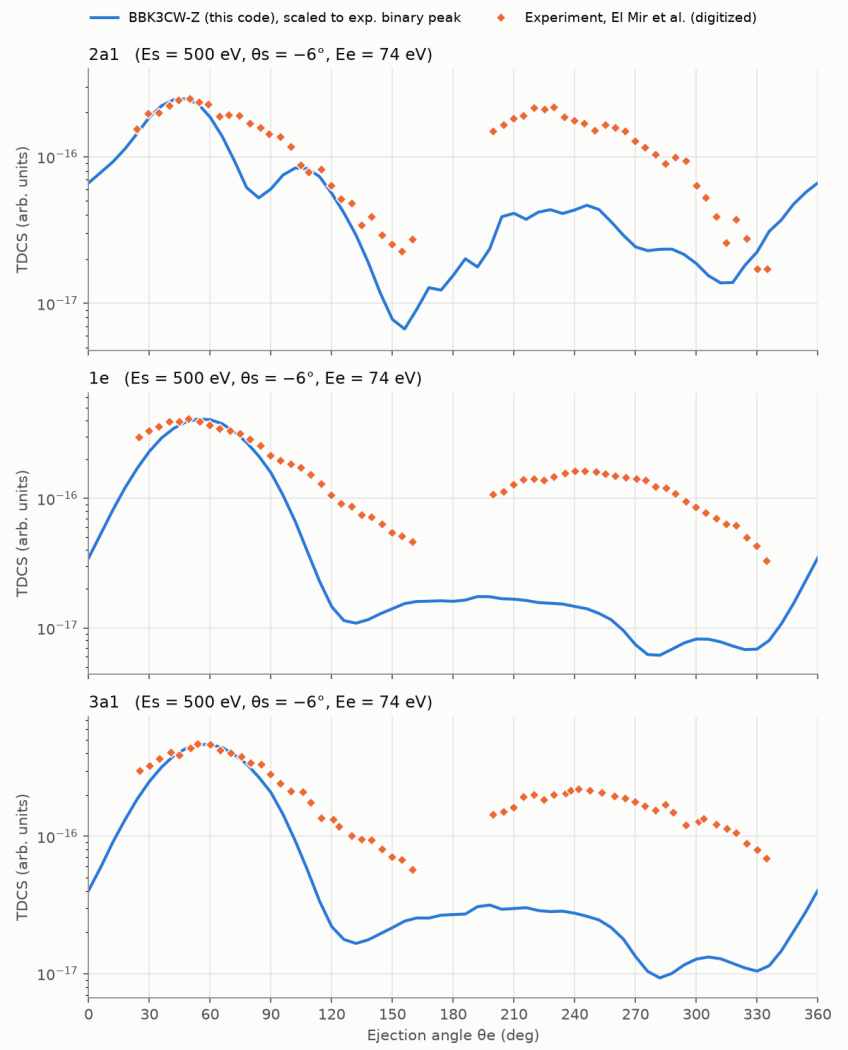


Figure 1. BBK3CW-Z (blue line) and the El Mir et al. experiment (orange diamonds) for $2a_1$, 1e and $3a_1$. Each model curve is scaled to the experimental binary maximum.

What works

- The **binary peak position** is close to experiment: 48–60° against 49–54°. The model is a few degrees too far forward for $3a_1$ and 1e.
- The **front side of the binary peak** (25–60°) follows the data well.

What fails

- **The recoil peak is almost missing.** For $3a_1$ and 1e the model is a flat plateau 5–10 times below the data, with no maximum near $-\mathbf{K}$. For $2a_1$ there is a weak bump at 246°, but it is still 4–5 times too low.
- **The binary peak is too narrow on its large-angle side.** At $\theta_e = 120°$, the 1e experiment is about 1×10^{-16} and the model about 1.5×10^{-17}, a factor of about 7; the same holds for $3a_1$. In the experiment the cross section falls slowly from the binary peak towards the recoil region; in the model it drops sharply after 90°.
- **$2a_1$ has a dip at about 80° and a second bump at 105°** that the experiment does not show (it has a smooth shoulder). $2a_1$ is mostly N 2s, whose momentum distribution has a node, and the model shows that node too sharply. In the real collision the distortion of the ejected electron would smear it out.

The first two failures have the same cause: in the model, the ejected electron is not scattered enough by the ion. Whatever leaves the binary direction (slow fall-off, valley filling, recoil peak) requires the ion to deflect the electron, and the BBK3CW-Z ejected wave barely does.

2. With the short-range term (isr = 1)

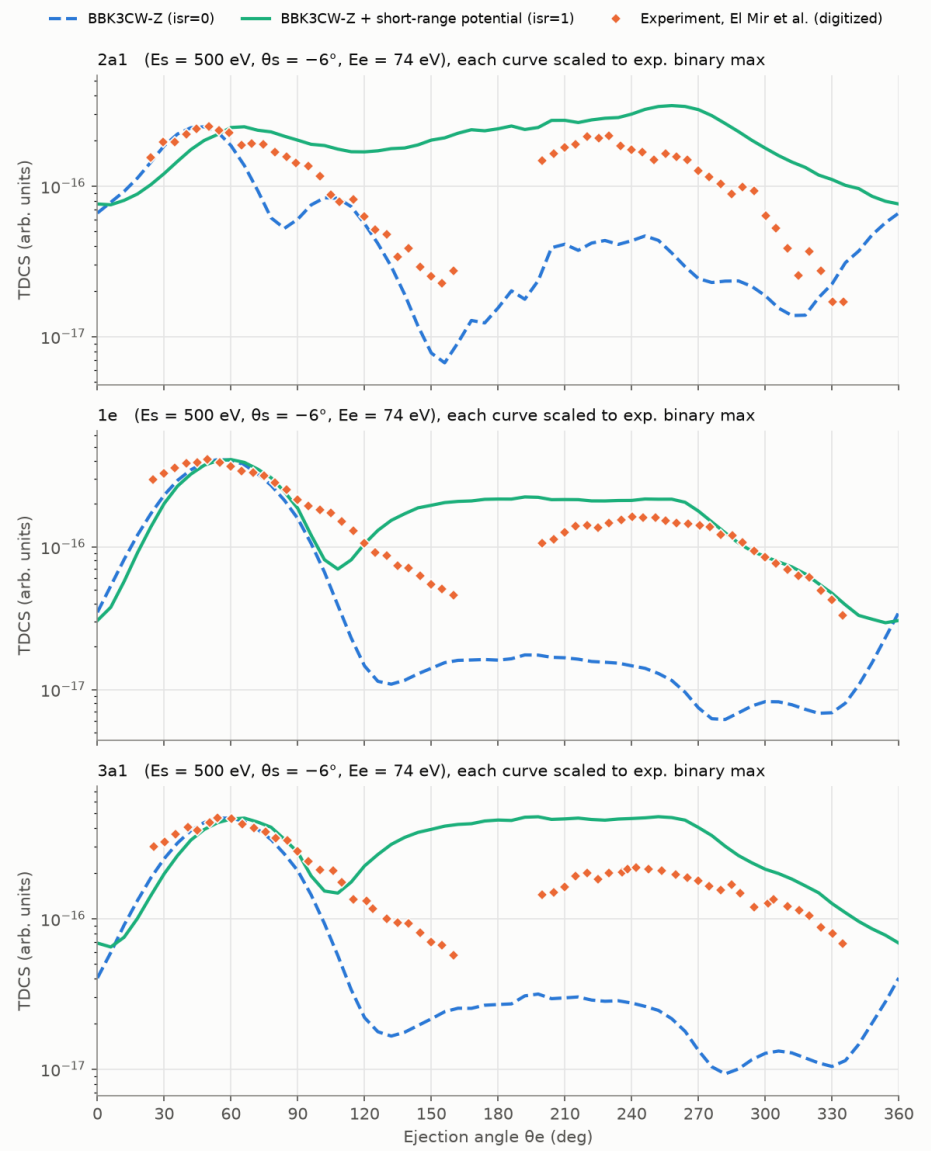


Figure 2. BBK3CW-Z (dashed blue), BBK3CW-Z with the short-range projectile–core potential $-(Z_{ion}(r)-1)/r$ (solid green), and experiment (orange diamonds).

This option adds the projectile–core attraction inside the molecule.

- **1e** is the best result so far: the binary peak is right, and the curve from 270° to 335° falls on the data. But 130–200° is filled by a plateau that is too high, so the recoil looks like a broad shelf instead of a peak centred at 240°.
- **$3a_1$**: the recoil is now about 2 times too high and much too broad; it starts right after the binary peak, at about 110°.
- **$2a_1$**: the binary peak loses its shape and shifts to 66°. The curve is almost flat from 60° to 270°, which is unphysical. $2a_1$ has the largest s part, so the large overlap between the ejected Coulomb wave and the orbital (0.37) inflates this term the most.
- **Binary peak position**: it moves forward to 60–66° in all three orbitals, away from the experimental 49–54°.

The term pushes in the right direction (it produces back-scattering) but in the wrong way: it adds a nearly isotropic contribution instead of a peak centred at $-\mathbf{K}$. That isotropic part is what one obtains when the ejected wave is not orthogonal to the orbital.

3. Orthogonalized ejected wave (iorth = 1)

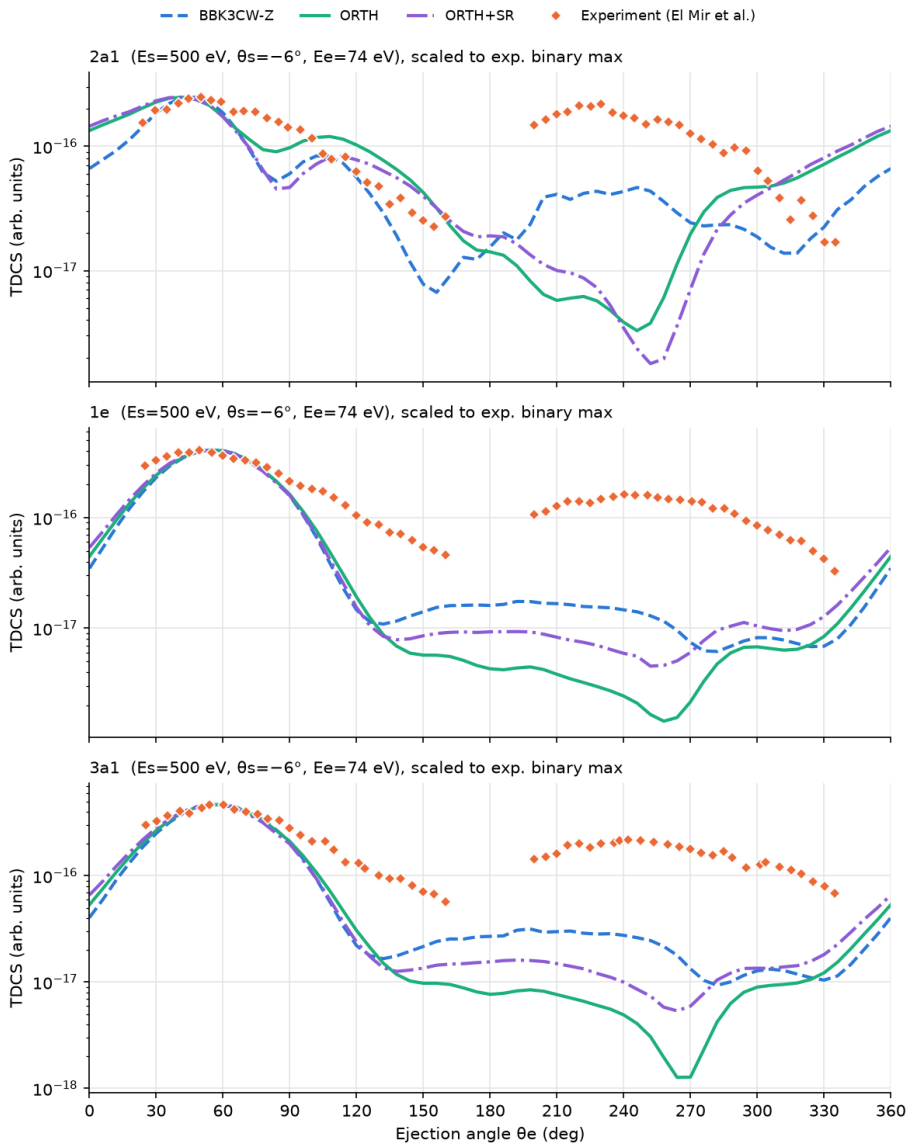


Figure 3. BBK3CW-Z (dashed blue), orthogonalized ejected wave (solid green), orthogonalized with the short-range term (dash-dot purple), and experiment (orange diamonds).

- **Binary peak**: unchanged and good (42–54°).
- **Recoil without the short-range term**: even lower than the original, with a deep minimum around 255–270°.
- **With the short-range term**: the isotropic shelf of Fig. 2 is gone, so $2a_1$ recovers its binary peak shape, but the recoil region falls back to about 10 times below the data.

Most of the extra recoil produced by the short-range term was therefore the overlap artefact; the physical part left is far too weak.

What the three figures say together

Observation	Meaning
Binary peak correct in all versions	The initial state (Moccia orbitals), the first-order projectile treatment and the kinematics are fine
Recoil and large-angle side 5–10 times low, for all 3 orbitals	A missing physical mechanism, not an orbital-specific problem
Short-range term raises it, but isotropically	That increase comes mainly from the non-orthogonal overlap, not from real back-scattering
Orthogonalization removes the increase	Once the artefact is gone, almost no real back-scattering is left in the ejected wave

All four observations point to the **ejected-electron wave**. It does not contain the strong back-scattering of the NH_3^+ core. Solving Eq. (11) of the theory report exactly with the same Z(r) gives large short-range phase shifts (2.6, 1.3 and 0.5 rad for l = 0, 1, 2) and an elastic $e^- + NH_3^+$ cross section 30–140 times the Rutherford (Z = 1) value at 120–180°. The variable-charge Coulomb wave of BBK3CW-Z keeps the Z = 1 phases and cannot reproduce this back-scattering. The proposed next step is therefore an exact distorted wave for the ejected electron (see theory_distorted_wave.md), not another adjustment of Z(r) or of the potential term.

Caveat. This is a shape comparison only. If El Mir et al. put the three orbitals on a common relative scale, the orbital-to-orbital ratios would give an additional test.